

JOAN WACHUKA MAINA

AI/ML Engineer | Full-Stack Developer

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[GitHub: github.com/jianna4](https://github.com/jianna4) | [Portfolio: portfolio-s2qb.vercel.app](https://portfolio-s2qb.vercel.app)

PROFESSIONAL SUMMARY

AI engineer who ships. Python • FastAPI • React • LangChain • LLMs. Built and deployed SheriaAI, a RAG legal chatbot for Kenya's Employment Act. Also built and contributed to RAG applications during internship at AFFAM Limited. Final-year Computer Science student with hands-on experience in AI/ML and a deep passion for building. Seeking AI internship at Mitsumi Distribution.

TECHNICAL SKILLS

Programming Languages:

Python, JavaScript, SQL

Frontend Development:

React.js, Tailwind CSS, HTML, CSS

Backend Development:

Django, FastAPI, REST APIs

AI & LLMs:

LangChain, OpenAI API, RAG, Vector Databases, Embeddings

Machine Learning & Data:

TensorFlow, Scikit-learn, Pandas, NumPy

Tools & Technologies:

Git, GitHub, Render, Vercel, Hugging Face

EDUCATION

Bachelor of Science in Computer Science

Murang'a University of Technology

2022 – Expected August 2026

- AI/ML Official – MUT Tech Community

- Member – Google Developer Student Clubs (GDSC)
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PROFESSIONAL EXPERIENCE

Software Developer Intern

AFFAM Limited

April 2025 – August 2025

- Developed and tested backend APIs using Django and FastAPI
 - Participated in system testing, debugging, and performance optimization
 - Contributed to development of retrieval-augmented generation (RAG) applications
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PROJECTS

SheriaAI – Kenya Employment Act Chatbot

Live Demo: sheria-ai-opal.vercel.app | *GitHub:* github.com/jianna4/sheria-ai

- Production-ready RAG (Retrieval-Augmented Generation) chatbot that answers questions about Kenya's Employment Act 2007, helping Kenyan employees know and understand their rights. Deployed with 203 document chunks processed.
- Implemented dual-mode system (Employee/Employer perspectives) – unique feature
- Built vector database from 70+ page legal document using LangChain + ChromaDB

SkinVision _ For detection and management of skin diseases affecting African Countries

Live Demo: jianna4-skin-disease-analyser.hf.space/ | *GitHub:* github.com/jianna4/skindisease-detector

- Developed a convolutional neural network (CNN) for skin disease classification
 - Performed data preprocessing, training, and evaluation
 - Analyzed model performance and misclassification patterns
 - Achieved First Runner-up in an JKUAT Inter-University AI Hackathon (2024)
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CERTIFICATIONS & ACHIEVEMENTS

- Cisco – Data Analytics Essentials
- FreeCodeCamp – Machine Learning with Python